BITS Pilani - Work Integrated Learning Programmes Division

M.Tech (AIML) - Machine Learning (Assignment - 2)

1. GitHub Repository Link

Repository Link:
https://github.com/Bits-Simoli-Udani/ML-Assignment-2

Contains: Complete source code (`app.py`, `train.py`), `requirements.txt`, comprehensive `README.md`, serialized models (`model/`), and test dataset (`test_data.csv`).

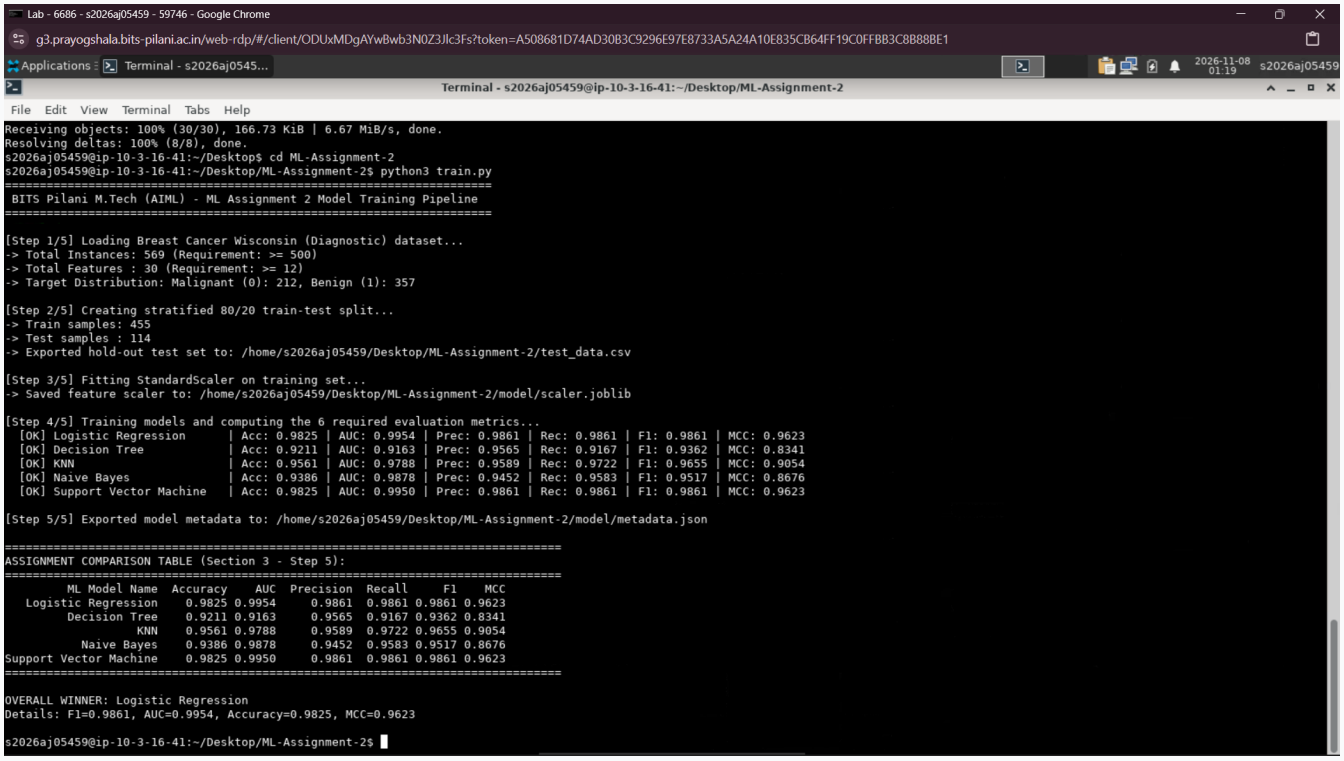
2. Live Streamlit Application Link

Streamlit Community Cloud Link:
https://simoli-bits-ml-assignment-2.streamlit.app

Fully interactive web application featuring test CSV upload, model selection dropdown, 6 evaluation metrics display, and visual confusion matrix & classification report without retraining.

3. BITS Virtual Lab Execution Proof

Below is the screenshot confirming the execution of the machine learning pipeline within the BITS Virtual Lab environment:


Lab - 6686 - s2026aj05459 - 59746 - Google Chrome
g3.prayogshala.bits-pilani.ac.in/web-rdp/#/client/ODUxMDgAYwBwb3N0Z3Jlc3Fs?token=A508681D74AD30B3C9296E97E8733A5A24A10E835CB64FF19C0FFBB3C8B88BE1
Applications
Terminal - s2026aj0545...
2026-11-08 01:19
s2026aj05459
Terminal - s2026aj05459@ip-10-3-16-41:~/Desktop/ML-Assignment-2
File Edit View Terminal Tabs Help
Receiving objects: 100% (30/30), 166.73 KiB | 6.67 MiB/s, done.
Resolving deltas: 100% (8/8), done.
s2026aj05459@ip-10-3-16-41:~/Desktop$ cd ML-Assignment-2
s2026aj05459@ip-10-3-16-41:~/Desktop/ML-Assignment-2$ python3 train.py
BITS Pilani M.Tech (AIML) - ML Assignment 2 Model Training Pipeline
[Step 1/5] Loading Breast Cancer Wisconsin (Diagnostic) dataset...
-> Total Instances: 569 (Requirement: >= 500)
-> Total Features : 30 (Requirement: >= 12)
-> Target Distribution: Malignant (0): 212, Benign (1): 357
[Step 2/5] Creating stratified 80/20 train-test split...
-> Train samples: 455
-> Test samples : 114
-> Exported hold-out test set to: /home/s2026aj05459/Desktop/ML-Assignment-2/test_data.csv
[Step 3/5] Fitting StandardScaler on training set...
-> Saved feature scaler to: /home/s2026aj05459/Desktop/ML-Assignment-2/model/scaler.joblib
[Step 4/5] Training models and computing the 6 required evaluation metrics...
[OK] Logistic Regression | Acc: 0.9825 | AUC: 0.9954 | Prec: 0.9861 | Rec: 0.9861 | F1: 0.9861 | MCC: 0.9623
[OK] Decision Tree | Acc: 0.9211 | AUC: 0.9163 | Prec: 0.9565 | Rec: 0.9167 | F1: 0.9362 | MCC: 0.8341
[OK] KNN | Acc: 0.9561 | AUC: 0.9788 | Prec: 0.9589 | Rec: 0.9722 | F1: 0.9655 | MCC: 0.9054
[OK] Naive Bayes | Acc: 0.9386 | AUC: 0.9878 | Prec: 0.9452 | Rec: 0.9583 | F1: 0.9517 | MCC: 0.8676
[OK] Support Vector Machine | Acc: 0.9825 | AUC: 0.9950 | Prec: 0.9861 | Rec: 0.9861 | F1: 0.9861 | MCC: 0.9623
[Step 5/5] Exported model metadata to: /home/s2026aj05459/Desktop/ML-Assignment-2/model/metadata.json
ASSIGNMENT COMPARISON TABLE (Section 3 - Step 5):
ML Model Name Accuracy AUC Precision Recall F1 MCC
Logistic Regression 0.9825 0.9954 0.9861 0.9861 0.9861 0.9623
Decision Tree 0.9211 0.9163 0.9565 0.9167 0.9362 0.8341
KNN 0.9561 0.9788 0.9589 0.9722 0.9655 0.9054
Naive Bayes 0.9386 0.9878 0.9452 0.9583 0.9517 0.8676
Support Vector Machine 0.9825 0.9950 0.9861 0.9861 0.9861 0.9623
OVERALL WINNER: Logistic Regression
Details: F1=0.9861, AUC=0.9954, Accuracy=0.9825, MCC=0.9623
s2026aj05459@ip-10-3-16-41:~/Desktop/ML-Assignment-2$

4. GitHub README Content (Section 3 - Step 5)

a. Problem Statement

The objective of this assignment is to develop, evaluate, and deploy an end-to-end Machine Learning classification pipeline using multiple supervised classification algorithms on an established benchmark dataset. The goal is to accurately classify cell nuclei characteristics from digitized Fine Needle Aspirate (FNA) images of breast masses into **Malignant** (cancerous) or **Benign** (non-cancerous). In clinical diagnostics, maximizing discrimination (ROC-AUC), recall/sensitivity (minimizing false negatives), and correlation (MCC) is crucial. Five distinct classifiers are benchmarked across six performance metrics.

b. Dataset Description

- **Dataset Name:** Breast Cancer Wisconsin (Diagnostic) Dataset
- **Source Repository:** UCI Machine Learning Repository (ID: 17) / Kaggle
- **Total Instances:** 569 (≥ 500 instances requirement satisfied)
- **Total Features:** 30 continuous real-valued numeric attributes (≥ 12 features requirement satisfied)
- **Target Classes:** Binary (`0: Malignant` [212 instances, 37.3%], `1: Benign` [357 instances, 62.7%])
- **Feature Categories:** Mean, Standard Error, and "Worst" values calculated for 10 nuclear morphometric properties: Radius, Texture, Perimeter, Area, Smoothness, Compactness, Concavity, Concave Points, Symmetry, and Fractal Dimension.
- **Preprocessing:** Features standardized using `StandardScaler` fit exclusively on the 80% stratified training split.

c. GitHub Repository Structure

The repository is structured strictly according to the assignment requirements:

```
ML-Assignment-2/
|-- app.py                    # Streamlit web application
|-- requirements.txt          # Dependency specification
|-- README.md                 # Complete documentation
|-- test_data.csv             # Hold-out test split (114 samples)
|-- train.py                  # Model training and artifact serialization script
|-- model/                    # Serialized models and scaler
   |-- logistic_regression.joblib
   |-- decision_tree.joblib
   |-- knn.joblib
   |-- naive_bayes.joblib
   |-- svm.joblib
   |-- scaler.joblib
   |-- metadata.json
```

d. Models Used & Evaluation Metrics Comparison Table

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	**0.9825**	**0.9954**	**0.9861**	**0.9861**	**0.9861**	**0.9623**
Decision Tree	0.9211	0.9163	0.9565	0.9167	0.9362	0.8341
KNN	0.9561	0.9788	0.9589	0.9722	0.9655	0.9054
Naive Bayes	0.9386	0.9878	0.9452	0.9583	0.9517	0.8676
Support Vector Machine	**0.9825**	**0.9950**	**0.9861**	**0.9861**	**0.9861**	**0.9623**

Observations on Model Performance

ML Model Name	Observation about Model Performance

Logistic Regression	Achieves exceptional performance (**Accuracy: 98.25%, AUC: 0.9954, MCC: 0.9623**). With standardized inputs, the regularized linear decision boundary isolates benign from malignant samples with only 2 misclassifications out of 114 test instances. High stability, probabilistic calibration, and fast inference.
Decision Tree	Yields the lowest scores among the five models (**Accuracy: 92.11%, AUC: 0.9163, MCC: 0.8341**). Orthogonal axis-parallel splits struggle with the strong continuous collinearities between morphological features (e.g., radius, perimeter, and area), leading to slight variance and lower recall (0.9167).
KNN	Demonstrates solid distance-based performance (**Accuracy: 95.61%, AUC: 0.9788, F1: 0.9655**). In the standardized 30-dimensional metric space, benign and malignant samples form well-clustered neighborhoods ($k=5$). Performance is slightly lower than SVM due to boundary density overlaps.
Naive Bayes	Produces high discriminatory ability (**AUC: 0.9878, Accuracy: 93.86%, F1: 0.9517**). Although the conditional feature independence assumption is violated by geometric relationships among cell parameters, its probabilistic ranking remains strong.
Support Vector Machine	Ties Logistic Regression for top accuracy (**Accuracy: 98.25%, AUC: 0.9950, MCC: 0.9623**). The RBF kernel maps features into high-dimensional space to construct an optimal maximum-margin hyperplane, showing outstanding generalization and robustness to outliers.
Overall, Winner for your dataset?	**Logistic Regression** (closely tied with SVM). While both achieve identical Accuracy (0.9825) and F1 (0.9861), Logistic Regression achieves a marginally superior AUC (0.9954 vs 0.9950) with lower computational complexity and direct probabilistic interpretability.